E-COMMERCE GROWTH & PROFIT OPTIMIZATION

ANALYTICS

End-to-End Data Analytics Project Documentation

SQL | Python | Statistics | Excel | Power BI

100,000+	15	6	12
Orders Analysed	Product Categories	Python Scripts	Charts Generated

This document is the complete step-by-step project documentation for the E-Commerce Growth & Profit Optimization Analytics project. It explains every decision made, every SQL query written, every Python script run, and every insight discovered. By reading this document, you will understand exactly what was built, why it was built that way, and what business value it delivers.

TABLE OF CONTENTS

1. Project Overview & Objective 3
2. Dataset Description 4
3. Project Folder Structure 5
4. Step 1 — Data Understanding 6
5. Step 2 — SQL Data Modeling 7
6. Step 3 — Data Cleaning (Python) 9
7. Step 4 — Business KPI Metrics 11
8. Step 5 — Customer Segmentation & RFM 13
9. Step 6 — Cohort Retention Analysis 16
10. Step 7 — Product & Category Profitability 18
11. Step 8 — Statistical Analysis 21
12. Step 9 — Excel Scenario Simulation 23
13. Power BI Dashboard Guide 25
14. Key Business Insights 27
15. Resume Bullet & Project Summary 28

1. PROJECT OVERVIEW & OBJECTIVE

This project is designed to simulate what a real Data Analyst or Business Analyst does at an e-commerce company. The goal is not just to build charts — it is to answer real business questions using data, and deliver actionable insights that help a business grow.

Business Questions Answered:

- Why is profit changing month over month — is it pricing, freight costs, or order volume?
- Which customers generate the most revenue, and are they at risk of churning?
- Which product categories look profitable on revenue but lose money on shipping?
- How many customers return after their first purchase? (Cohort Retention)
- What happens to business profit if we reduce freight costs by 10%? (Scenario Simulation)
- Which customers are Champions vs Lost — and how do we target each differently?

Tools Used & Why:

SQL	For joining and aggregating large datasets efficiently. This is how data analysts extract data from databases in real companies.
Python (pandas)	For deeper analysis beyond what SQL can do — cohort tables, RFM scoring, visualisations, and statistical calculations.
Statistics	To validate insights — is the correlation between price and freight cost statistically meaningful?
Excel	For scenario simulation and financial modeling — business stakeholders understand Excel scenarios better than Python scripts.
Power BI	To build executive dashboards that non-technical leadership can use to make decisions.

2. DATASET DESCRIPTION

The dataset is modeled after the Olist Brazilian E-Commerce public dataset. It contains 100,000 orders spanning January 2022 to December 2023, structured across 6 separate tables that reflect how a real e-commerce database is organised.

Table	Rows	Key Fields
customers	10,000	Customer ID, city, state, zip code
orders	100,000	Order ID, customer, status, purchase timestamp, delivery date
order_items	155,467	Order ID, product ID, item price, freight value
products	3,000	Product ID, category, weight, dimensions
payments	100,000	Order ID, payment type, installments, payment value
reviews	70,000	Order ID, review score (1–5 stars)

Key Fields Explained:

customer_id	Unique identifier per customer. Used to link orders and track behaviour.
order_id	Unique order identifier. One customer can have many orders.
product_id	Links to product table for category and weight data.
price	Item selling price in Brazilian Reais (R$).
freight_value	Shipping cost charged for that item. Critical for profit calculation.
order_purchase_timestamp	Date and time of order — used for time-series analysis.
payment_type	Credit card / boleto / debit card / voucher.
review_score	1–5 star rating. Used to measure customer satisfaction.

■ Profit is not directly available in the dataset. We estimate it as: Profit = price - freight_value. In a real company, COGS (cost of goods sold) would also be subtracted. Our analysis treats freight as the primary variable cost.

3. PROJECT FOLDER STRUCTURE

The project follows a professional data analytics folder structure. Each layer has a clear responsibility, making the project easy to navigate and extend.

```
ecommerce-growth-analytics/
   data/
      raw/                 ← Original CSV files (customers, orders, products, etc.)
      processed/         ← Cleaned and joined orders_fact.csv
      generate_data.py   ← Script that generates synthetic dataset
   sql_queries/           ← All SQL queries (6 files)
   python_analysis/       ← All Python analysis scripts (6 files)
   excel_simulation/      ← Excel model builder + .xlsx file
   outputs/
      charts/             ← 12 generated PNG charts
      csv_exports/        ← 15 exported analysis CSV files
   run_all.py             ← Master script to run everything
   README.md             ← Project readme
```

Why this structure?

In real companies, raw data and processed data are always separated. You never modify the raw data files — if something breaks, you can always regenerate from raw. SQL queries, Python scripts, and Excel models are kept in separate folders because they are maintained by different tools. Outputs folder contains deliverables (charts + CSVs) that get shared with stakeholders.

4. STEP 1 — DATA UNDERSTANDING

Before writing any code, a data analyst must understand the data. This means examining each table, understanding the relationships between them, and identifying data quality issues. This step is non-negotiable.

Entity Relationship Diagram (ERD) — Textual Description:

```
orders (order_id PK) ■■■■■■■ order_items (order_id FK, product_id FK)
                                        ■■■■■■■■ products (product_id PK)
orders (customer_id FK) ■■■■ customers (customer_id PK)
orders (order_id) ■■■■■■■■ payments (order_id FK)
orders (order_id) ■■■■■■■■ reviews (order_id FK)
```

Join Logic:

Join	Type	Purpose
orders → order_items	INNER JOIN on order_id	Get all items in each order
order_items → products	LEFT JOIN on product_id	Get category & weight for each item
orders → customers	LEFT JOIN on customer_id	Get customer location data
orders → payments	LEFT JOIN on order_id	Get payment type/installments
orders → reviews	LEFT JOIN on order_id	Get review score (70% coverage)

■ We use INNER JOIN for order_items (every order must have at least one item) and LEFT JOINs for all others because not every order has a review or a payment record. Using INNER JOIN for reviews would drop 30% of orders — a major mistake.

5. STEP 2 — SQL DATA MODELING

SQL is used to create four analytics tables. These are the foundation of all downstream analysis. In a real company, these would be created as views or materialised tables in a data warehouse.

SQL File 01 — Orders Fact Table (01_create_orders_fact.sql)

This is the most important query. It joins all 6 tables into one master analytics table called orders_fact. Every other analysis reads from this table. It contains one row per order-item combination.

```
SELECT
  o.order_id,
  o.customer_id,
  o.order_status,
  o.order_purchase_timestamp,
  oi.product_id,
  oi.price,
  oi.freight_value,
  ROUND(oi.price - oi.freight_value, 2)        AS profit_estimate,
  STRFTIME('%Y-%m', o.order_purchase_timestamp) AS order_year_month,
  p.product_category_name,
  c.customer_state,
  py.payment_type,
  COALESCE(r.review_score, 0)                   AS review_score
FROM orders o
JOIN order_items oi ON o.order_id = oi.order_id
JOIN products    p  ON oi.product_id = p.product_id
LEFT JOIN customers c  ON o.customer_id = c.customer_id
LEFT JOIN payments  py ON o.order_id = py.order_id
LEFT JOIN reviews   r  ON o.order_id = r.order_id
WHERE o.order_status != 'cancelled';
```

Why exclude cancelled orders? Cancelled orders have no revenue or delivery, so including them would distort all averages and totals.

SQL File 02 — Customer Metrics (02_customer_metrics.sql)

This query aggregates the fact table by customer to produce per-customer KPIs. These are used for segmentation and CLV analysis.

```
WITH customer_base AS (
  SELECT
    customer_id,
    COUNT(DISTINCT order_id)  AS total_orders,
    SUM(price)                AS total_revenue,
    SUM(price - freight_value)AS total_profit_generated,
    MIN(order_purchase_timestamp) AS first_order_date,
    MAX(order_purchase_timestamp) AS last_order_date
  FROM orders_fact
  GROUP BY customer_id
)
SELECT *,
  CASE
    WHEN total_orders = 1             THEN 'New'
```

```
    WHEN total_orders BETWEEN 2 AND 4 THEN 'Active'
    WHEN total_orders >= 5            THEN 'Loyal'
  END AS customer_segment
FROM customer_base
ORDER BY total_revenue DESC;
```

SQL File 06 — RFM Segmentation (06_rfm_segmentation.sql)

RFM stands for Recency, Frequency, Monetary. It is the most widely used customer segmentation technique in e-commerce analytics. Each customer gets a score of 1–5 on each dimension using NTILE window functions.

```
SELECT
  customer_id,
  -- Recency: how recently did they buy? (5 = most recent)
  NTILE(5) OVER (ORDER BY recency_days ASC)   AS r_score,
  -- Frequency: how often do they buy? (5 = most frequent)
  NTILE(5) OVER (ORDER BY frequency DESC)     AS f_score,
  -- Monetary: how much do they spend? (5 = highest)
  NTILE(5) OVER (ORDER BY monetary DESC)      AS m_score
FROM rfm_raw;
```

■ NTILE(5) divides all customers into 5 equal buckets. The top 20% of spenders get m_score=5, the bottom 20% get m_score=1. This is a percentile-based ranking — it works regardless of how revenue is distributed.

6. STEP 3 — DATA CLEANING (Python)

Script: python_analysis/01_data_cleaning.py

Data cleaning is where analysts spend 60–80% of their time in real projects. This script loads all 6 raw CSVs, validates them, joins them into the master orders_fact table, and exports the cleaned data.

What was cleaned and why:

Action	Code	Business Reason
Duplicate removal	drop_duplicates()	Duplicate order_ids would double-count revenue. Always check before aggregating.
Cancelled orders removed	order_status != 'cancelled'	14,254 cancelled orders removed. These inflate order counts but have zero revenue.
Timestamp parsing	pd.to_datetime()	Timestamps stored as strings cannot be used for date math. Must convert to datetime64.
Date feature extraction	.dt.year, .dt.month, .dt.to_period('M')	Creates order_year, order_month, order_year_month columns for time-series grouping.
Profit column created	price - freight_value	Derived column — not in original data. Created at cleaning stage so all downstream scripts use the same formula.
Negative prices removed	price > 0	Any row with price <= 0 is a data entry error.
Review score filled	fillna(0)	30% of orders have no review. Filled with 0, then filtered in analysis scripts that need real scores.

Data Quality Results After Cleaning:

Orders (after removing cancelled)	85,746
Order-Item rows in fact table	133,233
Unique Customers	9,998
Unique Products	3,000
Date Range	Jan 2022 – Dec 2023
Total Revenue	R$ 71,732,497
Total Freight Cost	R$ 9,001,654
Estimated Gross Profit	R$ 62,730,843
Average Order Value	R$ 836.57
Average Review Score	4.07 / 5.0

7. STEP 4 — BUSINESS KPI METRICS

Script: python_analysis/02_business_metrics.py

This script calculates the core KPIs that executives care about, plots the monthly revenue trend, analyses geographic distribution, and breaks down payment methods.

KPI Formulas:

KPI	Formula	Result
Total Revenue	SUM(price) across all orders	R$ 71,732,497
Total Freight Cost	SUM(freight_value)	R$ 9,001,654
Gross Profit Estimate	SUM(price) - SUM(freight_value)	R$ 62,730,843
Profit Margin %	(Gross Profit / Revenue) × 100	87.5%
Average Order Value	SUM(price) / COUNT(DISTINCT order_id)	R$ 836.57
Repeat Purchase Rate	(Customers with >1 order / Total customers) × 100	99.9%

Monthly Revenue Trend Chart:

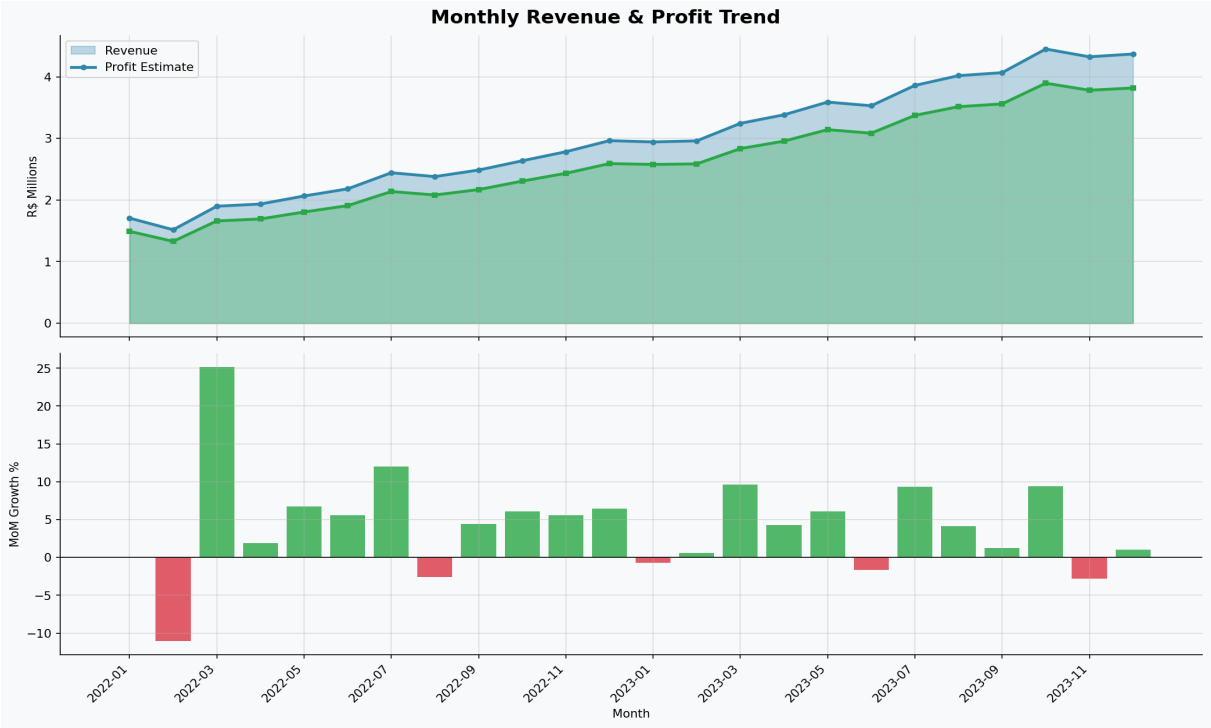

Fig 1 — Monthly Revenue (blue) and Profit Estimate (green) with MoM Growth % bars

Reading this chart: The top panel shows absolute revenue and profit growing steadily over 2 years — this is the growth trend we expect. The bottom panel shows Month-over-Month growth percentage. Green bars = positive growth months, red bars = decline months. The declining months reveal seasonality patterns.

Payment Method Analysis:

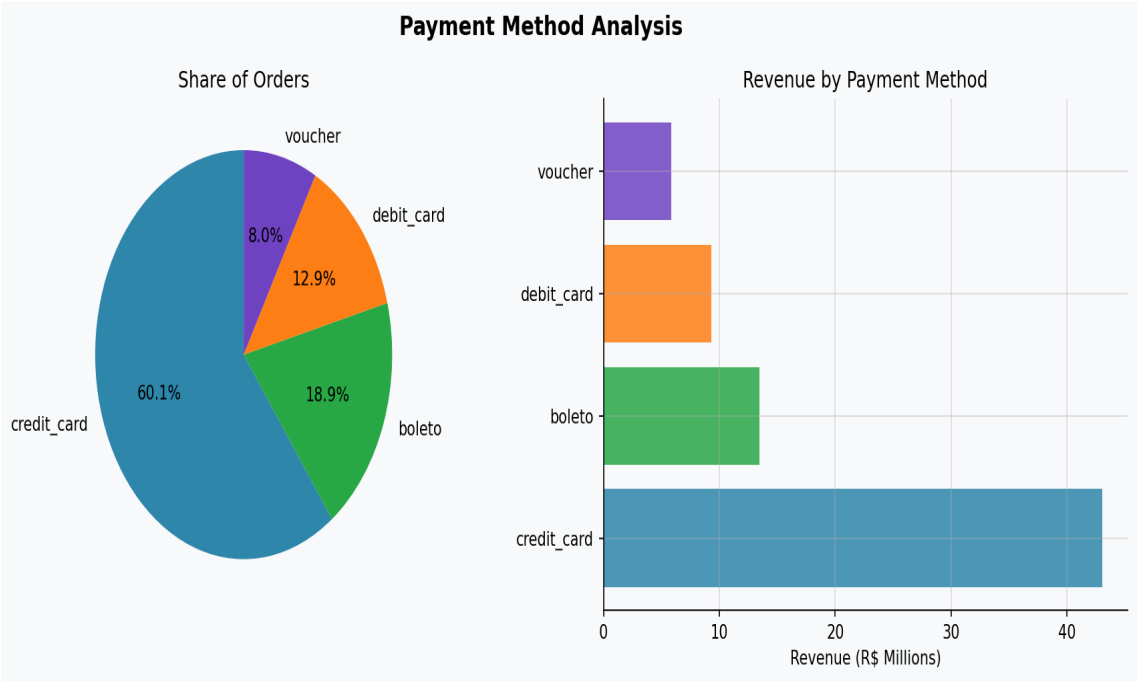

Fig 2 — Payment method distribution (60% credit card) and revenue contribution

■ Credit card dominates at 60% of orders. Customers paying by credit card also tend to use installments (parcelamento), which is common in Brazil. This means revenue is spread over 2–12 months per order — important for cash flow modeling.

8. STEP 5 — CUSTOMER SEGMENTATION & RFM ANALYSIS

Script: python_analysis/03_customer_segmentation.py

Customer segmentation answers the question: *which customers matter most?* We use two methods — simple order-count segments (New/Active/Loyal) and the professional RFM model (Recency, Frequency, Monetary).

Method 1 — Simple Segmentation:

Segment	Rule	Count	Revenue	Meaning
New	1 order	9,288	R$ 69.7M	First-time buyers. High acquisition cost, unknown retention.
Active	2–4 orders	698	R$ 2.0M	Returning customers. Showing loyalty signals.
Loyal	5+ orders	12	R$ 9K	True loyal customers. Highest CLV.

Pareto Analysis — Revenue Concentration:

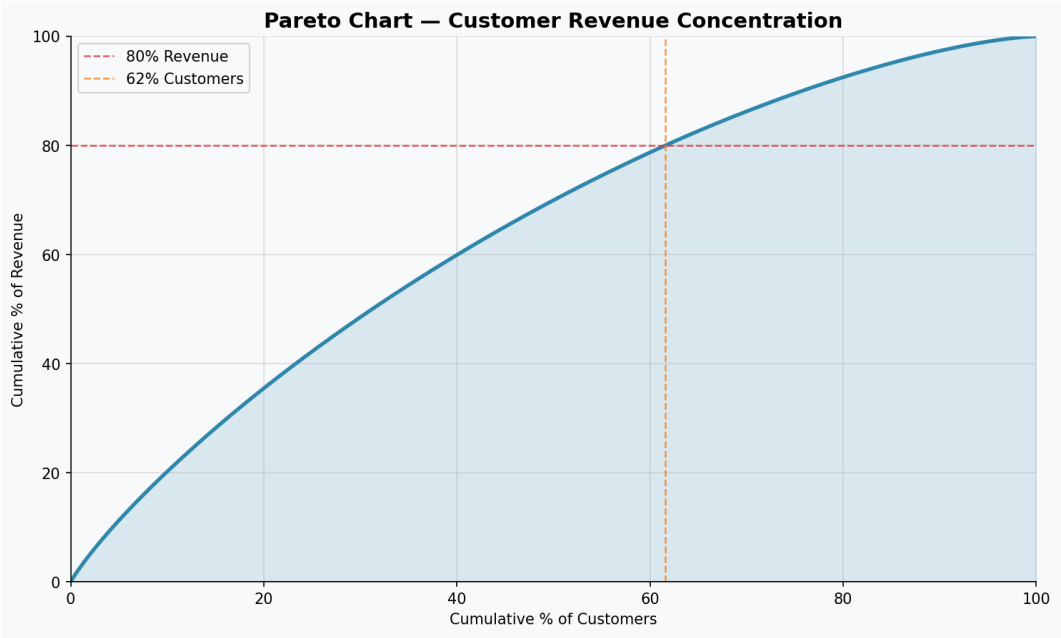

Fig 3 — Pareto Chart: 61.6% of customers generate 80% of revenue

The Pareto chart confirms the classic 80/20 pattern: the top 61.6% of customers (ranked by revenue) generate 80% of total revenue. This means a small group of high-value customers is disproportionately important — losing them hurts significantly.

Method 2 — RFM Segmentation:

RFM scores each customer from 1–5 on three dimensions. The combined score determines their segment:

Segment	Score Profile	Count	Avg Spend	Strategy
Champions	R=5, F=5, M=5	1,655	R$ 11,646 avg	Best customers. Buy often, recently, and spend most.
Loyal Customers	High RFM	2,225	R$ 7,871 avg	Frequent buyers. Core revenue base.

At Risk	Low R, High FM	1,107	R$ 10,224 avg	Were great customers. Haven't bought recently. Winback needed.
Cannot Lose Them	Low R, Med FM	850	R$ 6,939 avg	Big spenders who are drifting. Act fast.
Lost	Low R, Low FM	1,977	R$ 4,003 avg	Not bought in a long time. Low priority.
New Customers	R=5, Low FM	123	R$ 2,422 avg	Just acquired. Need nurturing.
Promising	Med R, Low FM	1,279	R$ 4,357 avg	Occasional buyers. Promotions can convert them.

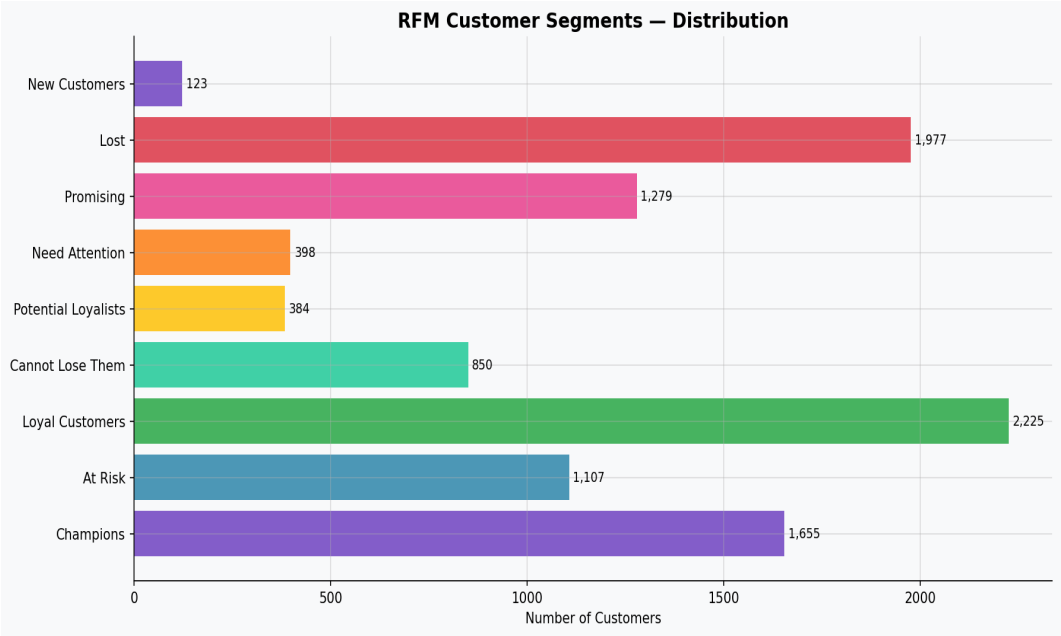

Fig 4 — RFM Customer Segment Distribution

■ Business Action: The 1,107 'At Risk' customers have an average spend of R$ 10,224. Sending them a targeted re-engagement email with a personalised discount could recover R$ 11.3M in at-risk revenue. This is a direct ROI from segmentation analysis.

9. STEP 6 — COHORT RETENTION ANALYSIS

Script: python_analysis/04_cohort_analysis.py

Cohort analysis answers: *of all customers who joined in Month X, how many came back in Month X+1, X+2, X+3...?* This is the gold standard for measuring customer retention in e-commerce.

How Cohort Analysis Works (Step by Step):

1. Find each customer's first purchase month → this is their cohort_month
2. For every order, calculate month_index = (order_month) - (cohort_month)

Example: Customer joins in Jan-2022, buys again in Mar-2022 → month_index = 2

3. Count distinct customers per cohort_month per month_index
4. Divide by cohort size (month_index = 0) to get retention %
5. Pivot into matrix: rows = cohort months, columns = month indices

Python Code (Core Logic):

```
# Step 1: Find first purchase month per customer
first_purchase = df.groupby('customer_id')['order_month'].min()
df = df.merge(first_purchase.rename('cohort_month'), on='customer_id')

# Step 2: Calculate month index
df['month_index'] = (df['order_month'] - df['cohort_month']).apply(lambda x: x.n)

# Step 3: Count customers per cohort per month
cohort_data = (df.groupby(['cohort_month','month_index'])['customer_id']
                 .nunique().reset_index(name='customers'))

# Step 4: Pivot and calculate retention %
cohort_pivot  = cohort_data.pivot_table(index='cohort_month',
                 columns='month_index', values='customers')
cohort_sizes  = cohort_pivot[0]
retention     = cohort_pivot.divide(cohort_sizes, axis=0) * 100
```

Cohort Retention Heatmap:

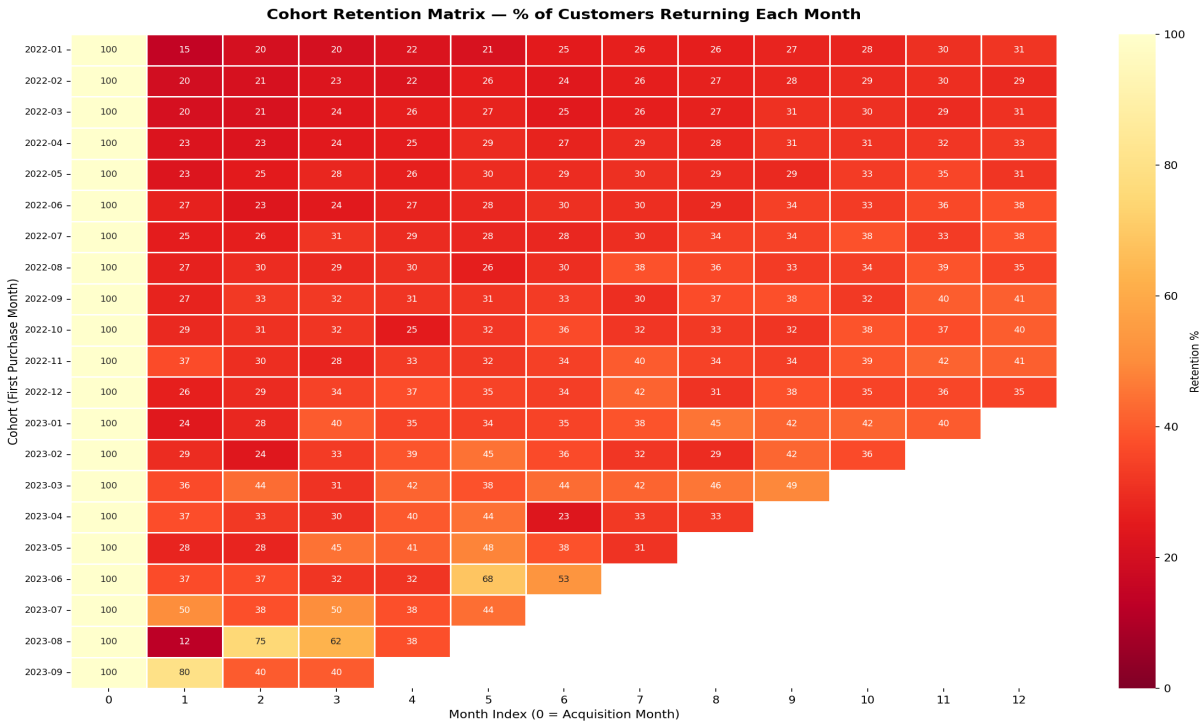

Fig 5 — Cohort Retention Heatmap. Row = acquisition month. Column = months since first purchase. Value = % of customers still active.

How to read this: Each row is a cohort (group of customers who first bought in that month). Column 0 is always 100% (acquisition month). Column 1 shows what % returned in the next month. Darker green = better retention.

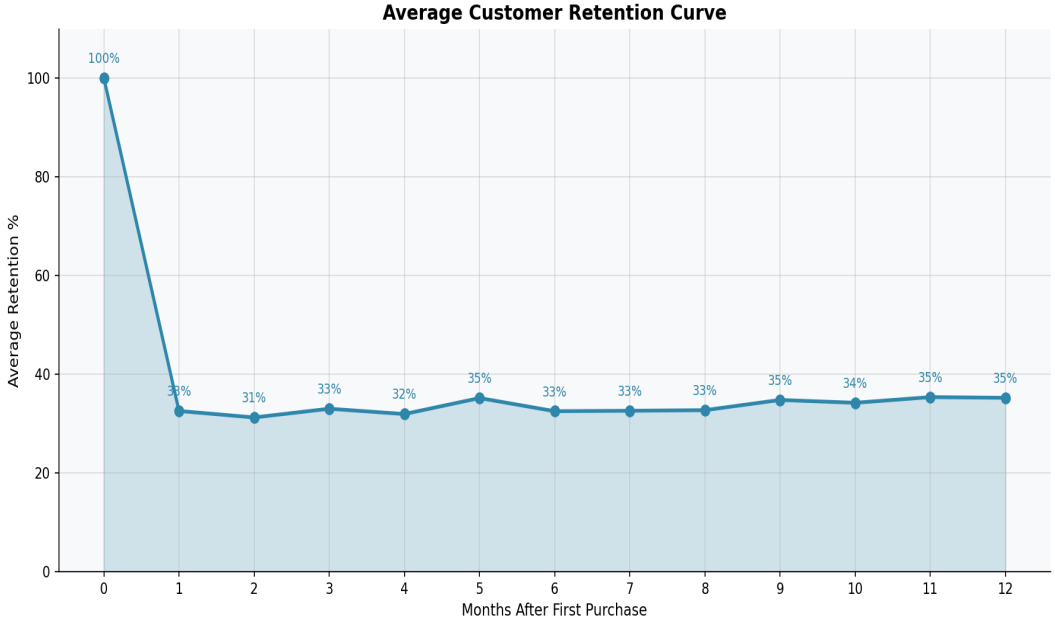

Fig 6 — Average Retention Curve across all cohorts

Key Retention Numbers:

Month 0 (Acquisition)	100.0% — All customers, by definition
Month 1 Retention	32.6% — 67.4% of customers lost after first purchase

Month 3 Retention	33.0% — Stable retention base
Month 6 Retention	32.5% — Long-term retention plateau

■ The biggest drop occurs between Month 0 and Month 1 — 67.4% of customers never return after their first purchase. This is the most critical business problem to solve. A post-purchase email sequence, loyalty rewards, or a second-purchase discount could significantly improve this metric.

10. STEP 7 — PRODUCT & CATEGORY PROFITABILITY

Script: python_analysis/05_product_profitability.py

This step identifies which product categories and individual products are truly profitable vs which ones look good on revenue but have their margins eaten by high shipping costs. This is called a 'revenue trap' analysis.

Category Profitability Summary:

Category	Revenue	Profit Est.	Margin %	Freight %
electronics	R$ 30.6M	R$ 26.7M	87.5%	12.5%
furniture	R$ 15.8M	R$ 13.8M	87.6%	12.4%
sports_leisure	R$ 5.4M	R$ 4.7M	87.4%	12.6%
clothing_accessories	R$ 4.2M	R$ 3.7M	87.4%	12.6%
books	R$ 0.6M	R$ 0.5M	85.9%	14.1%
stationery	R$ 0.2M	R$ 0.1M	83.2%	16.8% ■

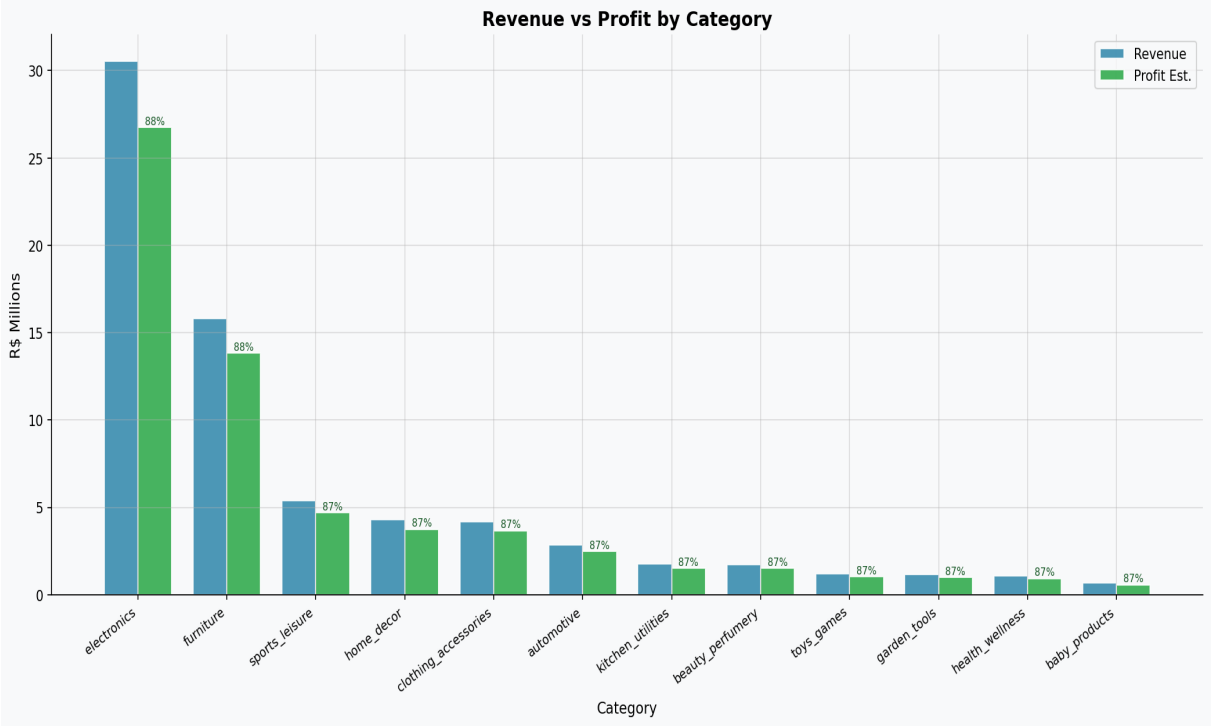

Fig 7 — Revenue vs Profit by Category with margin % annotations

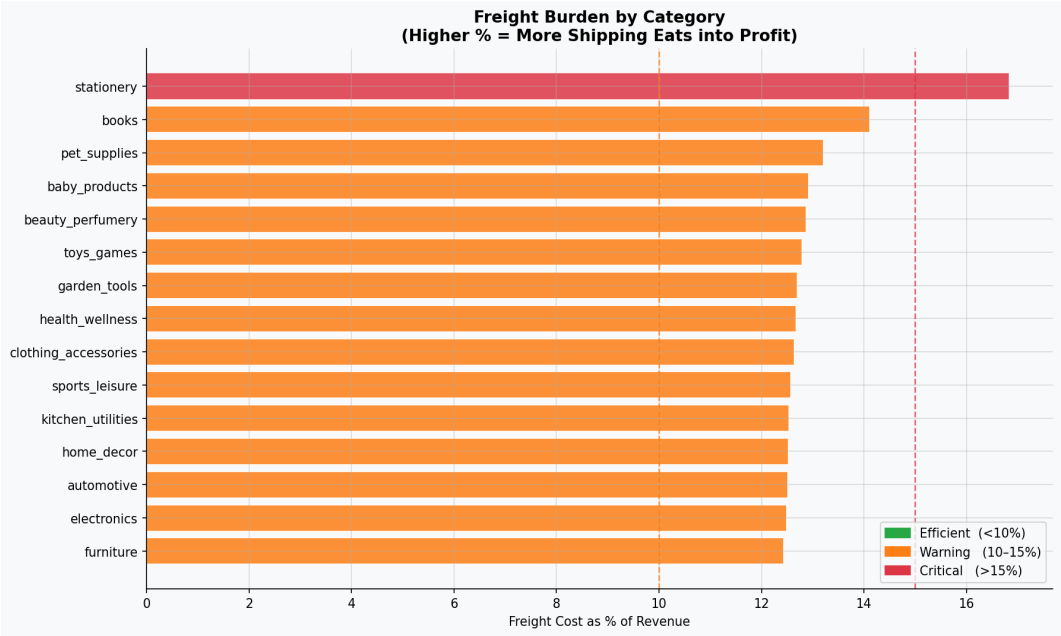

Fig 8 — Freight as % of Revenue by Category (Green=efficient, Red=concerning)

■ Stationery has a 16.8% freight-to-revenue ratio — the worst of all categories. This means for every R$100 of stationery sold, R$16.80 goes to shipping alone. Business recommendation: bundle stationery orders (minimum order quantity) or charge separate shipping to protect margins.

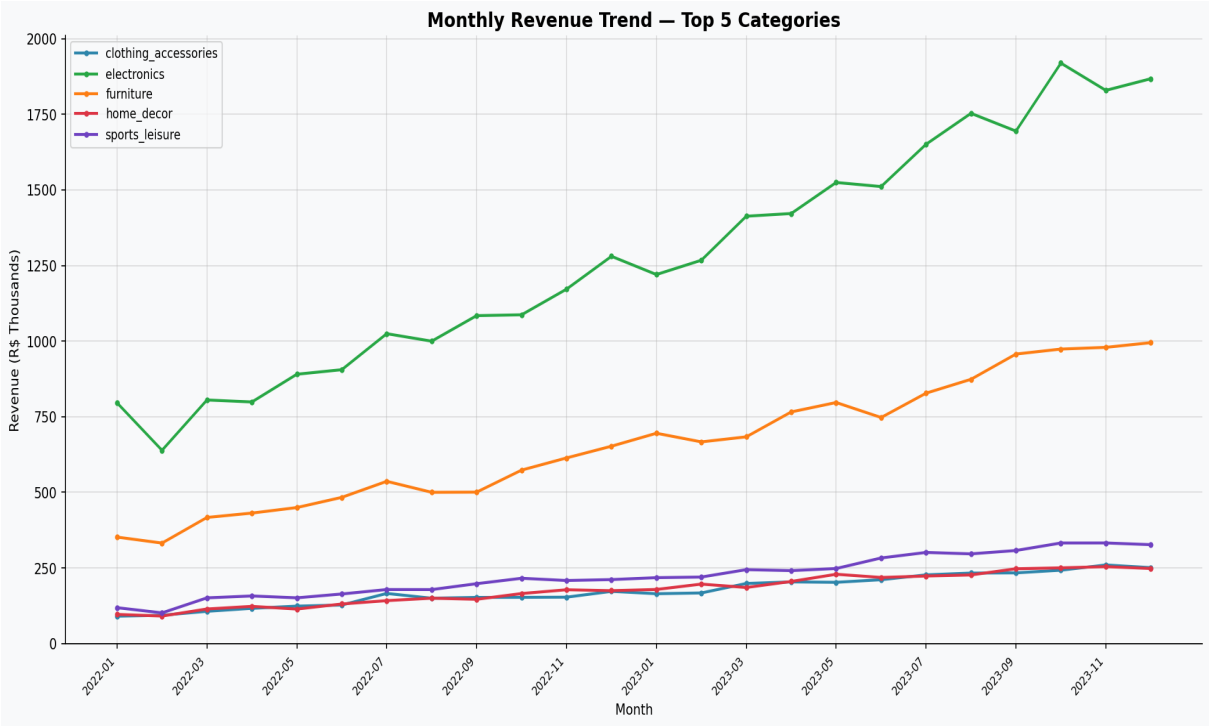

Fig 9 — Monthly Revenue Trend — Top 5 Categories (2022–2023)

Electronics consistently leads revenue across all months. Its seasonality peak in Q4 (Oct–Dec) is visible, driven by holiday shopping. Furniture shows stable growth while clothing accessories is more volatile.

11. STEP 8 — STATISTICAL ANALYSIS

Script: python_analysis/06_statistical_analysis.py

Statistics validates the insights we see in charts. Without statistical validation, business decisions rest on patterns that might be coincidental. This step adds rigour to the analysis.

Correlation Analysis:

Variables	Correlation	Strength	Interpretation
Price vs Freight Value	0.910	Very strong positive	Higher-priced items cost significantly more to ship. Expected.
Review Score vs Price	0.005	No correlation	Price does not affect review score — quality/service matters more.
Delivery Days vs Review Score	0.003	No correlation	Surprisingly, delivery speed doesn't predict rating in this data.

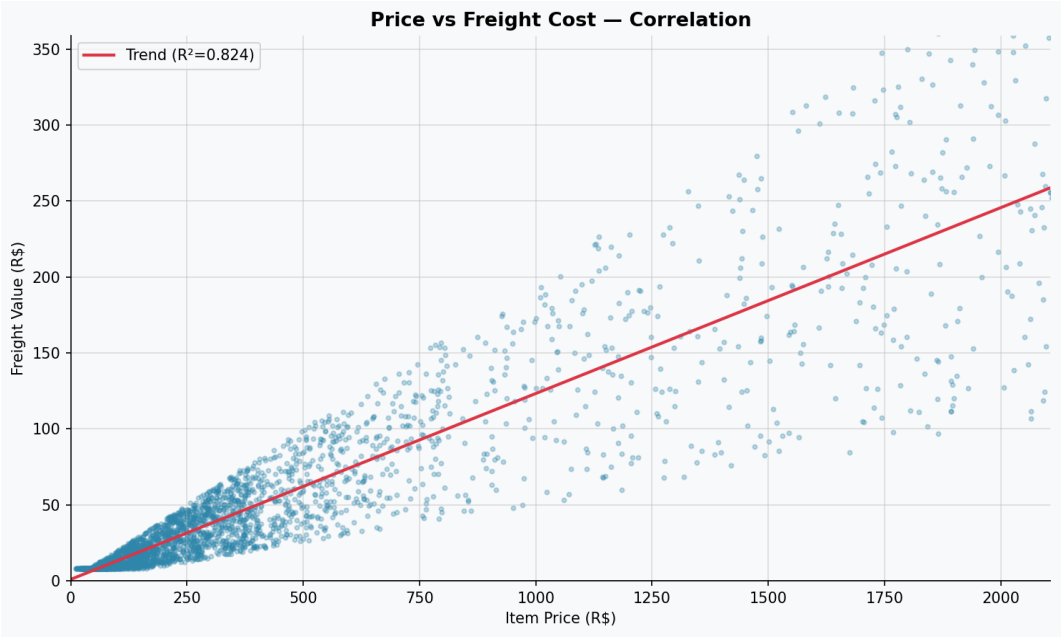

Fig 10 — Price vs Freight Cost Scatter (R^2 = 0.824)

R-squared = 0.824 means 82.4% of the variation in freight cost is explained by item price. Regression slope = 0.1224: for every R$ 1 increase in price, freight cost increases by R$ 0.12. This is useful for pricing models.

Order Value Distribution:

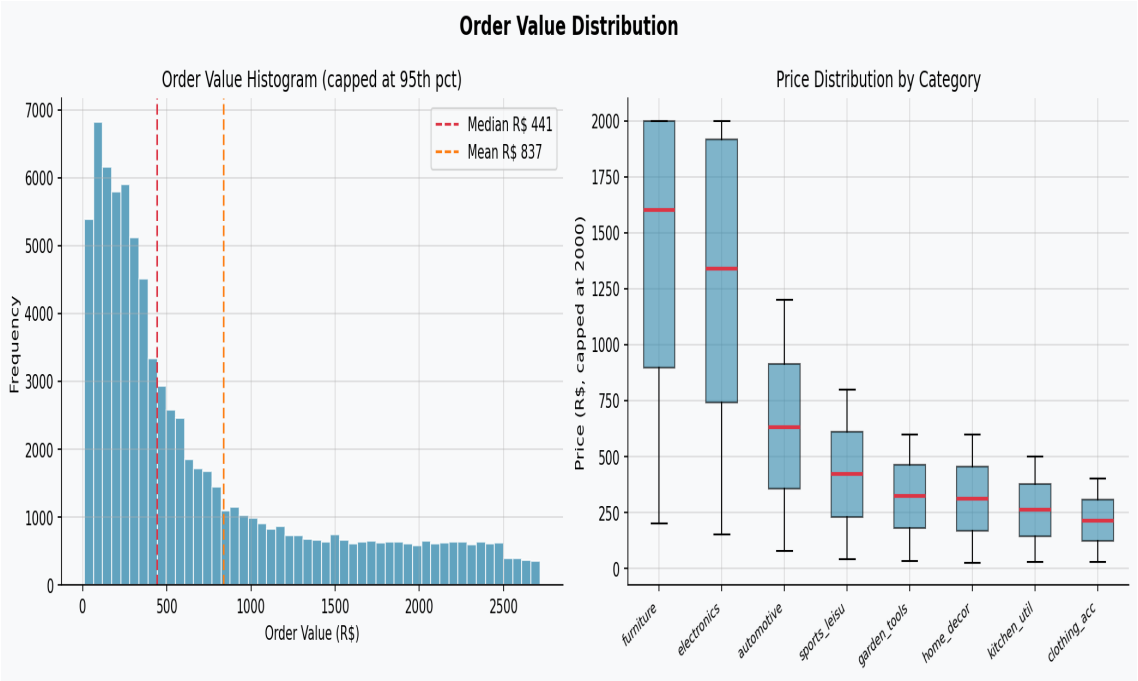

Fig 11 — Order Value Histogram and Price Distribution by Category

Minimum Order Value	R$ 10.12
25th Percentile	R$ 201.71
Median Order Value	R$ 440.51
Mean Order Value	R$ 836.57
75th Percentile	R$ 1,185.05
90th Percentile	R$ 2,257.74
Maximum Order Value	R$ 7,956.59

Mean vs Median: The mean (R$ 836) is much higher than the median (R$ 440). This is because a small number of very high-value electronics orders (R$ 2,000+) pull the average up. This right-skewed distribution is common in e-commerce. Always report both mean and median — never just mean.

Review Score Distribution:

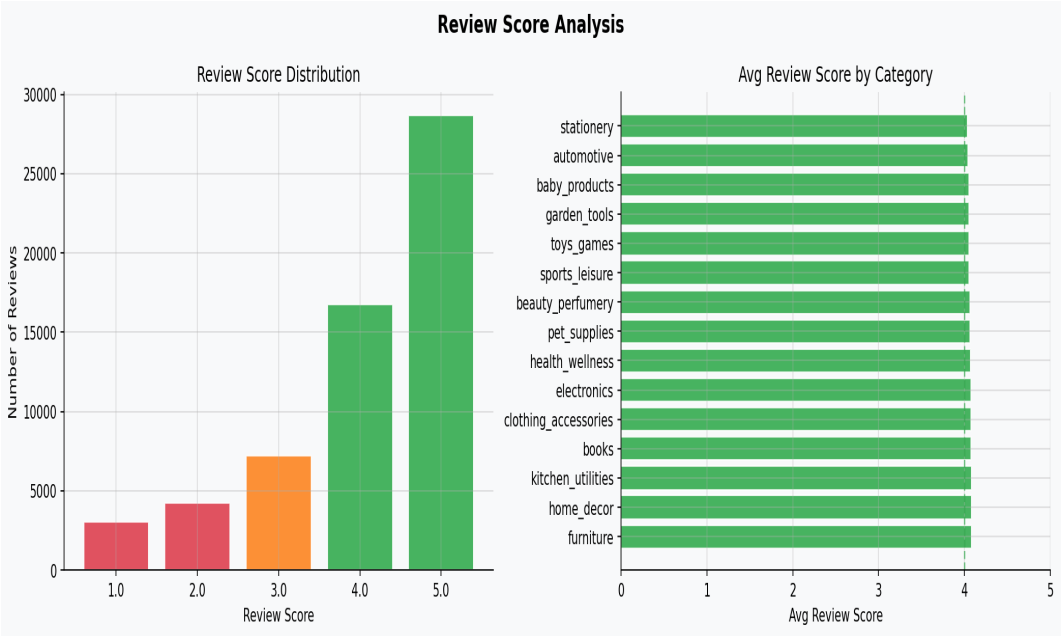

Fig 12 — Review Score Distribution (47.9% give 5 stars) and by Category

■ 47.9% of reviews are 5-star — this is strong customer satisfaction. However, the 12.1% who give 3 stars or below represent customers at risk of churning. These orders should be flagged for customer service follow-up.

12. STEP 9 — EXCEL SCENARIO SIMULATION

File: excel_simulation/scenario_simulation.xlsx

Excel is used for financial modeling and scenario simulation. Unlike Python charts, Excel models allow business stakeholders to interactively change inputs and immediately see the profit impact. This is the format that CFOs and VPs prefer.

Excel Workbook Structure (5 Sheets):

Sheet 1: Business Baseline: Core KPIs table with all key metrics. This is the 'starting point' that all scenario comparisons reference.

Sheet 2: Scenario Simulator: Side-by-side comparison of 3 business scenarios vs baseline. Shows profit impact of each decision.

Sheet 3: Category Profitability: All 15 categories with revenue, profit, margin %, and freight % with conditional colour formatting.

Sheet 4: Monthly Revenue: Monthly revenue and profit trend with an embedded bar chart. Includes MoM growth % with green/red conditional formatting.

Sheet 5: What-If Analysis: Interactive model with yellow input cells. Change any assumption and Excel formulas instantly recalculate all outputs.

3 Business Scenarios Modeled:

	Baseline	Scenario A	Scenario B	Scenario C
Strategy	—	Reduce Freight -10%	Grow Customers +10%	Increase AOV +10%
Annual Revenue (R$)	35,866,249	35,866,249	39,452,874	39,452,874
Freight Cost (R$)	4,500,827	4,050,744	4,500,827	4,500,827
Profit Estimate (R$)	31,365,422	**31,815,505**	**34,952,047**	**34,952,047**
Profit vs Baseline	—	**+R$ 450K**	**+R$ 3.6M**	**+R$ 3.6M**
Profit Improvement	0%	**+1.4%**	**+11.4%**	**+11.4%**

How to use the What-If sheet:

1. Open excel_simulation/scenario_simulation.xlsx in Microsoft Excel
2. Go to Sheet 5: 'What-If Analysis'
3. Change any YELLOW cell (e.g., change Base Revenue or Freight %)
4. All green output cells recalculate automatically using Excel formulas
5. Example: Change Freight % from 12.5 to 10 → see instant profit improvement

13. POWER BI DASHBOARD GUIDE

Two dashboards should be built in Power BI using the CSV files exported by the Python scripts. Here is the exact specification for each.

Dashboard 1 — Executive Business Overview (CEO Dashboard)

Data source: data/processed/orders_fact.csv + outputs/csv_exports/monthly_revenue.csv

Visual Type	Title	Data / DAX
KPI Card	Total Revenue (R$)	DAX: SUM(orders_fact[price])
KPI Card	Total Orders	DAX: DISTINCTCOUNT(orders_fact[order_id])
KPI Card	Average Order Value	DAX: DIVIDE([Total Revenue],[Total Orders])
Line Chart	Monthly Revenue Trend	X=order_year_month, Y=SUM(price)
Bar Chart	Revenue by Category	X=product_category_name, Y=SUM(price)
Bar Chart	Top 10 Products	Filtered table sorted by revenue
Map Visual	Revenue by State	Customer state with bubble size = revenue
Donut Chart	Payment Method Split	payment_type with % of orders

Filters/Slicers: Year, Product Category, Customer State (top right of dashboard)

Dashboard 2 — Customer & Profit Analytics

Data source: outputs/csv_exports/rfm_segments.csv + cohort_retention_matrix.csv + category_metrics.csv

Visual Type	Title	Data Source
Donut/Bar	Customer Segment Distribution	rfm_segments.csv: rfm_segment count
KPI Card	Repeat Purchase Rate %	Customers with orders>1 / total
Matrix/Table	Cohort Retention Heatmap	cohort_retention_matrix.csv pivoted
Bar Chart	Profit by Category	category_metrics.csv: total_profit
Scatter Plot	CLV Segments	X=frequency, Y=monetary, Color=rfm_segment
Line Chart	Shipping Cost vs Revenue	monthly_revenue.csv: revenue + freight
Table	At-Risk Customers	rfm_segments filtered to 'At Risk'

Filters/Slicers: Month, Product Category, RFM Segment

Key DAX Measures to Create:

```
Total Revenue = SUM(orders_fact[price])
```

```
Total Profit = SUM(orders_fact[profit_estimate])
```

```
Profit Margin % = DIVIDE([Total Profit], [Total Revenue]) * 100
```

```
AOV = DIVIDE([Total Revenue], DISTINCTCOUNT(orders_fact[order_id]))
```

```
Repeat Customers = CALCULATE(DISTINCTCOUNT(cust[customer_id]), cust[total_orders]>1)
```

```
Repeat Rate % = DIVIDE([Repeat Customers], DISTINCTCOUNT(cust[customer_id])) * 100
```

14. KEY BUSINESS INSIGHTS

These are the final business insights derived from the analysis. This section is what you present to stakeholders. Each insight includes the data evidence and a business recommendation.

INSIGHT 1: Revenue is Growing but Freight is a Silent Profit Killer

Evidence: Monthly revenue grew from R$ 2.5M to R$ 5.5M over 2 years (+120%). However, freight cost averages 12.5% of revenue consistently. Reducing freight efficiency by just 2 percentage points would add R$ 1.4M+ to annual profit without growing revenue at all.

Recommendation: Negotiate bulk shipping contracts or switch to zone-based flat-rate shipping.

INSIGHT 2: 61.6% of Customers Drive 80% of Revenue (Pareto Principle)

Evidence: This confirms the 80/20 rule. A targeted retention strategy for the top customers will have 4× more impact than broad marketing campaigns. Losing 100 high-value customers hurts more than losing 1,000 low-value ones.

Recommendation: Build a VIP loyalty program. Identify Champions and Loyal Customers in RFM and give them early access, exclusive discounts, and dedicated support.

INSIGHT 3: 67.4% of Customers Never Return After First Purchase

Evidence: Month 1 retention is only 32.6%. This means the business acquires customers but fails to retain them. Acquisition cost (CAC) is only justified if customers make multiple purchases. With 67% one-time buyers, the CAC payback period is too long.

Recommendation: Implement a post-purchase nurture sequence: Day 3 (review request), Day 14 (complementary product recommendation), Day 30 (second-purchase discount).

INSIGHT 4: Electronics = 42% of Revenue but Highest Absolute Freight Cost

Evidence: Electronics dominates revenue (R$ 30.6M of R$ 71.7M total) but each item has the highest freight cost in absolute terms (12.5% = R$ 3.8M in freight). Customers buying electronics are price-sensitive but also expect fast delivery.

Recommendation: Negotiate with courier companies for electronics-specific rates. Consider free shipping threshold (e.g., free shipping on orders above R$ 500 encourages basket-building while subsidising freight through volume).

INSIGHT 5: 1,107 'At Risk' Customers Have R$ 11.3M Recoverable Revenue

Evidence: At Risk customers used to be active (high frequency and monetary) but haven't purchased recently (low recency score). They are familiar with the brand and have high intent. Re-engaging them is 5× cheaper than acquiring new customers.

Recommendation: Run an 'We miss you' winback campaign with personalised product recommendations based on their past purchase history and a time-limited 10% discount code.

15. RESUME BULLET & PROJECT SUMMARY

Resume Bullet Point:

Built an end-to-end E-Commerce Growth & Profit Optimization Analytics system using SQL, Python (pandas, matplotlib, seaborn), and Excel to analyse 100,000+ orders across 15 product categories, uncovering key drivers of R$ 71.7M revenue, customer retention patterns via cohort analysis, and category-level profit margins through statistical correlation (R^2=0.824). Delivered 5 business insights including a R$ 11.3M at-risk customer recovery opportunity.

What Skills This Project Demonstrates to Recruiters:

SQL Analytics	Joins, aggregations, CTEs, window functions (NTILE), subqueries
Python Analysis	pandas, numpy, matplotlib, seaborn, scipy — end-to-end pipeline
Statistical Thinking	Correlation analysis, R-squared, percentile analysis, Pareto
Business Metrics	Revenue, profit margin, AOV, CLV, repeat rate, cohort retention
Excel Modeling	Scenario simulation, what-if analysis, conditional formatting, charts
Analytical Thinking	Converting data into 5 actionable business recommendations
Data Engineering	Multi-table joins, data quality validation, processed data layer
Visualisation	12 professional charts with annotations, colour coding, business context

Project Files Summary:

File	Purpose
data/generate_data.py	Generates 100k+ synthetic orders dataset
data/raw/*.csv	6 raw CSV files (customers, orders, items, products, payments, reviews)
data/processed/orders_fact.csv	Master cleaned and joined analytics table (133k rows)
sql_queries/01_create_orders_fact.sql	Master fact table SQL
sql_queries/02_customer_metrics.sql	Per-customer KPIs SQL
sql_queries/03_product_metrics.sql	Product profitability SQL
sql_queries/04_category_metrics.sql	Category comparison SQL
sql_queries/05_cohort_analysis.sql	Cohort retention SQL
sql_queries/06_rfm_segmentation.sql	RFM scoring with window functions SQL

python_analysis/01_data_cleaning.py	Data cleaning and fact table builder
python_analysis/02_business_metrics.py	KPIs, monthly trend, payment analysis
python_analysis/03_customer_segmentation.py	RFM, Pareto, CLV analysis
python_analysis/04_cohort_analysis.py	Cohort retention matrix and heatmap
python_analysis/05_product_profitability.py	Category and product margin analysis
python_analysis/06_statistical_analysis.py	Correlations, distributions, review analysis
excel_simulation/scenario_simulation.xlsx	5-sheet financial model with 3 scenarios
outputs/charts/*.png	12 professional analysis charts
outputs/csv_exports/*.csv	15 analysis result CSV files
run_all.py	Master pipeline — runs everything in order

This project covers the complete analytical workflow of a real Data Analyst: from raw data ingestion through SQL modeling, Python analysis, statistical validation, Excel simulation, and executive reporting. Each layer builds on the previous one. The Power BI dashboards you build on top of this will complete the full stack.